



DG INTERNS HUB

Cybersecurity Internship

WEEK 3 ASSIGNMENT

ADVANCED CYBERSECURITY PRACTICAL TASK

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The assignment guideline requires a 12-15 page PDF covering these sections. This report follows that structure and uses actual lab findings.

1. Introduction

This report documents a basic security assessment of an authorized virtual lab. The work combines network discovery, Nmap scanning, Wireshark traffic analysis, vulnerability/security review, firewall hardening, and post-hardening verification. The workflow follows a security analyst approach: collect evidence, interpret it carefully, apply defensive controls, and verify the final state.

Testing was limited to the user's own virtual lab. Public third-party systems were not used as assessment targets.

2. Objectives

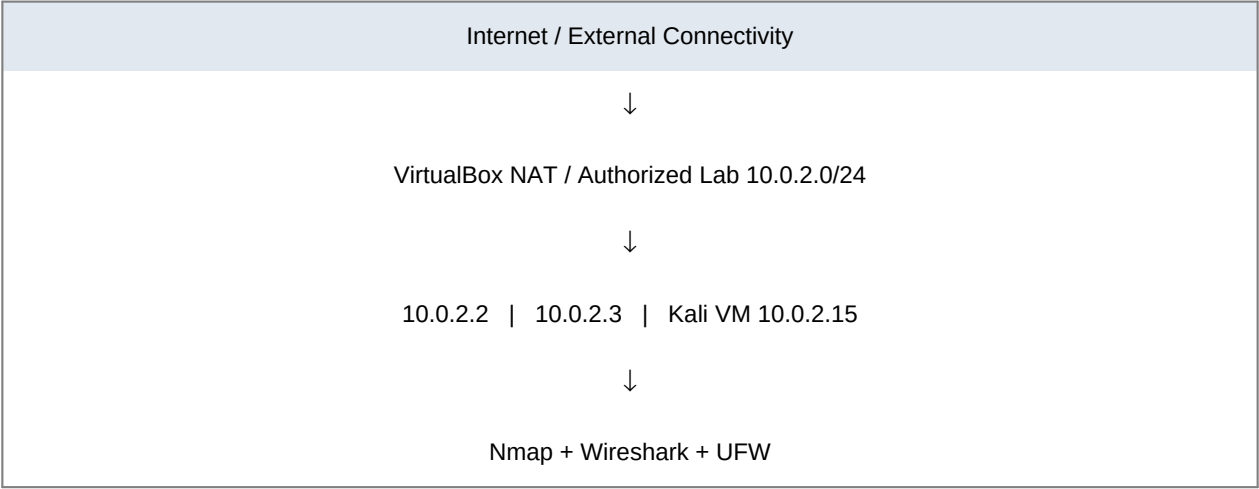
The objectives were to identify active lab devices, inspect exposed services, observe common protocols, correlate Nmap behavior with packet captures, identify evidence-based security findings, apply at least three hardening improvements, and document before/after results.

Evidence collected included network configuration, host discovery, service scans, ICMP and DNS captures, TCP SYN behavior, UFW rules, and final port verification.

3. Lab Environment

The assessment used a Kali Linux virtual machine in VirtualBox. The active interface was **eth0**, and the Kali VM used **10.0.2.15/24**. Host discovery observed **10.0.2.2**, **10.0.2.3**, and **10.0.2.15**.

4. Network Diagram



The diagram intentionally contains no personal device names, credentials, public account identifiers, or precise location information.

5. Tools Used

Tool	Purpose
Kali Linux	Security testing environment
Nmap	Host discovery and port/service scanning
Wireshark	Packet capture and protocol analysis
VirtualBox	Authorized isolated virtual lab
UFW	Host firewall hardening

6. Nmap Methodology

The workflow began with interface configuration, followed by host discovery on the authorized 10.0.2.0/24 subnet. A service/version scan was then performed against the Kali VM. Post-hardening verification specifically checked TCP ports 21 and 23.

Commands: ip a; sudo nmap -sn 10.0.2.0/24; sudo nmap -sV 10.0.2.15; sudo nmap -p 21,23 10.0.2.15.

7. Nmap Results

```
[user prompt redacted] UP> mtu 65536 qdisc noqueue state UNKNOWN group def
autl qlen 1000
link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
inet 127.0.0.1/8 scope host lo
    valid_lft forever preferred_lft forever
inet6 ::1/128 scope host noprefixroute
    valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP g
roup default qlen 1000
link/ether 08:00:27:99:54:7f brd ff:ff:ff:ff:ff:ff
inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
    valid_lft 80846sec preferred_lft 80846sec
inet6 fd00::ec22:cd82:1d26:8c39/64 scope global temporary dynamic
    valid_lft 84997sec preferred_lft 12997sec
inet6 fd00::a00:27ff:fe99:547f/64 scope global dynamic mngtmpaddr noprefi
xroute
    valid_lft 84997sec preferred_lft 12997sec
inet6 fe80::a00:27ff:fe99:547f/64 scope link noprefixroute
    valid_lft forever preferred_lft forever
```

Figure 1. Network configuration evidence.

```
[user prompt redacted] 10/24
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-09-18 22:21 IST
Nmap scan report for 10.0.2.2
Host is up (0.00058s latency).
MAC Address: 52:55:0A:00:02:02 (Unknown)
Nmap scan report for 10.0.2.3
Host is up (0.00072s latency).
MAC Address: 52:55:0A:00:02:03 (Unknown)
Nmap scan report for 10.0.2.15
Host is up.
Nmap done: 256 IP addresses (3 hosts up) scanned in 2.27 seconds
```

Figure 2. Nmap host discovery: three active lab hosts.

7. Nmap Results - Continued

```
[user prompt redacted] 10.0.2.15 TCP 60 80 → 53670 [S]
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-09-18 22:23 IST → 54660 [S]
Nmap scan report for 10.0.2.15 17.151.222 TCP 74 39482 → 80 [S]
Host is up (0.000032s latency). 15 TCP 60 80 → 39482 [S]
All 1000 scanned ports on 10.0.2.15 are in ignored states.
Not shown: 1000 closed tcp ports (reset)

Service detection performed. Please report any incorrect results at https://n
map.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 0.68 seconds
```

Figure 3. Service/version scan of 10.0.2.15.

The discovery scan reported three active hosts. The service scan against 10.0.2.15 reported the host as up while all 1000 commonly scanned TCP ports were closed. Therefore, no listening TCP service was identified in that scan.

8. Advanced Nmap Analysis

Because the tested host returned the scanned TCP ports as closed, there was no exposed TCP service version for Nmap to enumerate. A later targeted scan of ports 21 (FTP) and 23 (Telnet) also reported both ports closed.

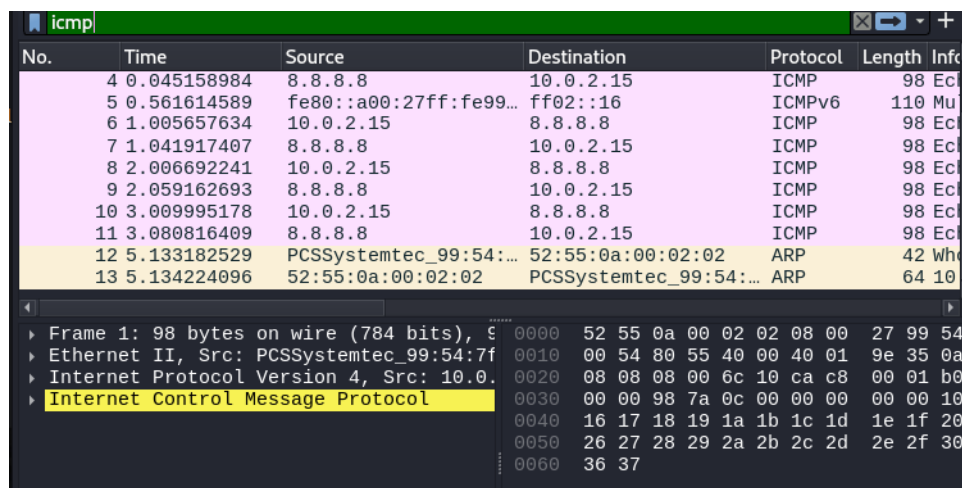
A closed result means no service was accepting a connection on that tested port at that moment. It should not be described as proof that the firewall alone closed the port.

9. Wireshark Methodology

Wireshark captured traffic on the Kali VM's eth0 interface. Controlled traffic was generated to make packet behavior easy to observe. Ping produced ICMP Echo traffic, nslookup produced DNS traffic, HTTP testing produced request/response traffic, and TCP filters were used to inspect connection establishment.

Display filters used during the practical included `icmp`, `dns`, `tcp`, `tcp.flags.syn == 1`, and `tcp.analysis.retransmission`.

10. Packet Analysis



The image shows a Wireshark packet capture window with the display filter set to 'icmp'. The packet list shows several ICMP Echo (ping) requests and replies. The first request is from 8.8.8.8 to 10.0.2.15 (packet 4). Subsequent packets show a bidirectional exchange between 10.0.2.15 and 8.8.8.8. The packet details pane for packet 1 shows the structure of an ICMP Echo request: Ethernet II, Internet Protocol Version 4, and Internet Control Message Protocol.

No.	Time	Source	Destination	Protocol	Length	Info
4	0.045158984	8.8.8.8	10.0.2.15	ICMP	98	Echo (ping) request
5	0.561614589	fe80::a00:27ff:fe99...	ff02::16	ICMPv6	110	Multicast
6	1.005657634	10.0.2.15	8.8.8.8	ICMP	98	Echo (ping) request
7	1.041917407	8.8.8.8	10.0.2.15	ICMP	98	Echo (ping) reply
8	2.006692241	10.0.2.15	8.8.8.8	ICMP	98	Echo (ping) request
9	2.059162693	8.8.8.8	10.0.2.15	ICMP	98	Echo (ping) reply
10	3.009995178	10.0.2.15	8.8.8.8	ICMP	98	Echo (ping) request
11	3.080816409	8.8.8.8	10.0.2.15	ICMP	98	Echo (ping) reply
12	5.133182529	PCSSystemtec_99:54:...	52:55:0a:00:02:02	ARP	42	Who-has (request)
13	5.134224096	52:55:0a:00:02:02	PCSSystemtec_99:54:...	ARP	64	Reply

Offset	0000	0010	0020	0030	0040	0050	0060
0	52	55	0a	00	02	02	08
8	00	27	99	54			
16	00	54	80	55	40	00	40
24	01	9e	35	0a			
32	08	08	08	00	6c	10	ca
40	c8	00	01	b0			
48	00	00	98	7a	0c	00	00
56	00	00	10				
64	16	17	18	19	1a	1b	1c
72	1d	1e	1f	20			
80	26	27	28	29	2a	2b	2c
88	2d	2e	2f	30			
96	36	37					

Figure 4. ICMP Echo requests and replies captured in Wireshark.

The ICMP capture showed bidirectional request/reply communication between 10.0.2.15 and 8.8.8.8, demonstrating successful reachability during the test.

10. Packet Analysis - DNS

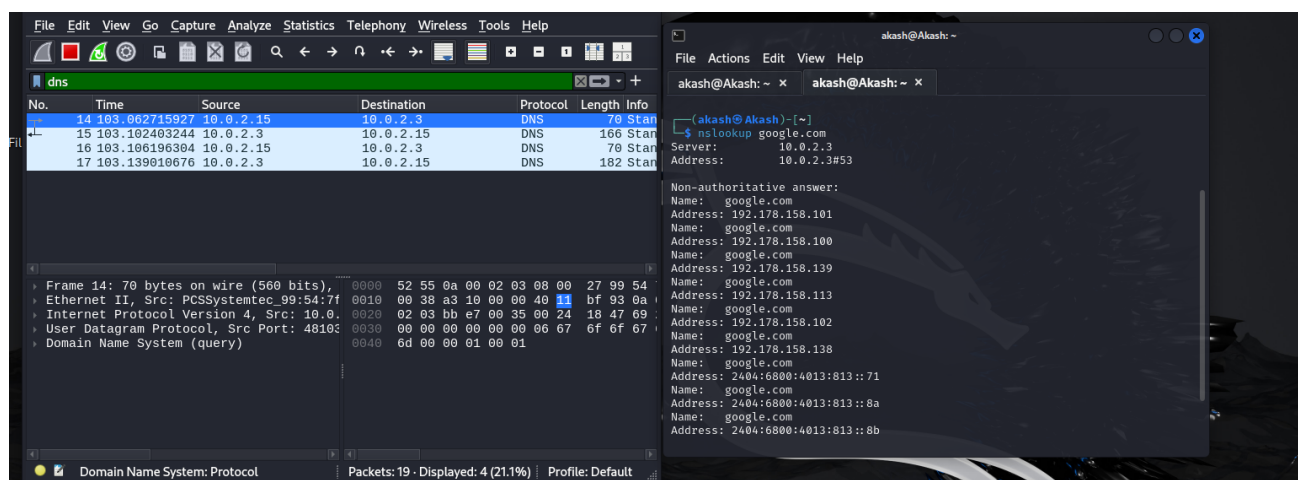


Figure 5. DNS query and response evidence for google.com.

DNS traffic showed queries from 10.0.2.15 to the lab DNS service at 10.0.2.3 and corresponding responses. Both IPv4 (A) and IPv6 (AAAA) resolution activity was visible.

This is normal name-resolution traffic. The evidence was not automatically labeled suspicious or malicious.

11. Nmap + Wireshark Investigation

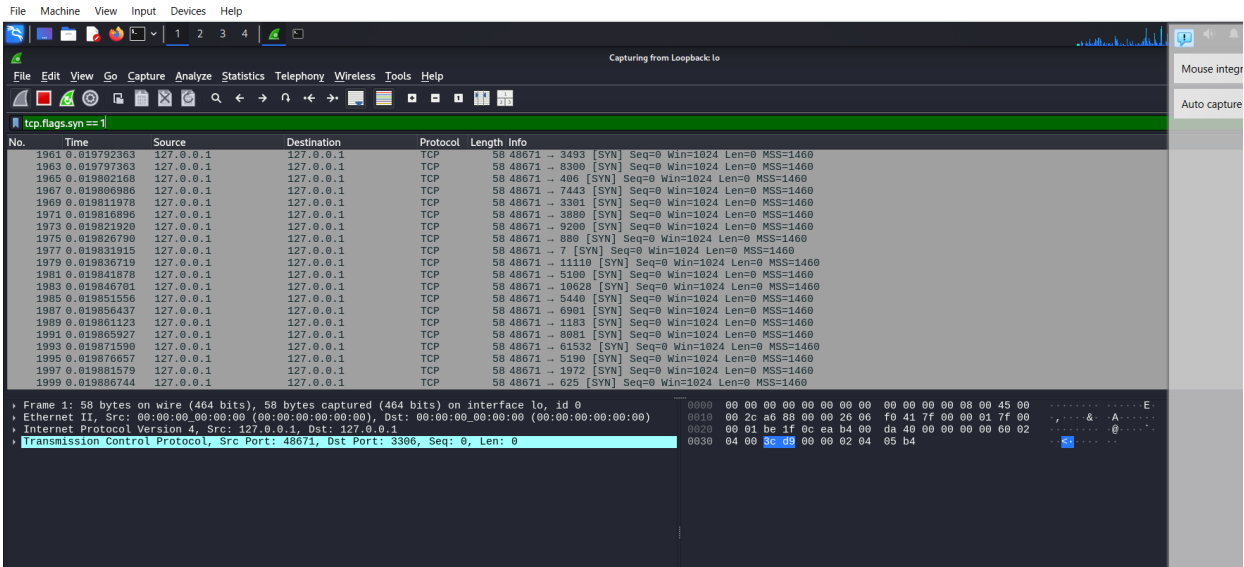


Figure 6. TCP SYN/SYN-ACK analysis evidence.

Network/Scan State	Packet Evidence	Interpretation
Reachable host	ICMP request/reply	Network path responds
Reachable TCP service	SYN -> SYN/ACK	Service appears reachable
Closed TCP port	SYN -> RST may occur	No listening service
No expected response	No matching reply	Filtering/loss may require investigation

Nmap and Wireshark complement each other: Nmap summarizes host/port state, while Wireshark shows the packets used to reach that conclusion.

12. Vulnerability Assessment

The available evidence did not identify a confirmed exploitable vulnerability or exposed versioned service on 10.0.2.15. Findings are therefore framed as configuration/exposure observations rather than overstated vulnerability claims.

13. Risk Classification

#	Finding	Evidence	Risk	Potential Impact	Recommendation
1	Firewall initially inactive	UFW inactive before enablement	Medium	Less host-level inbound control	Enable/retain UFW
2	No explicit FTP deny rule initially	15/tcp rule added	Low	Future FTP exposure could add attack surface	Keep FTP blocked unless needed
3	No explicit Telnet deny rule initially	23/tcp rule added	Medium	Telnet would be plaintext if enabled	Keep Telnet blocked
4	No exposed TCP service detected	100 TCP ports closed	Low	Current TCP exposure limited	Continue periodic scans
5	Normal DNS/ICMP visible	Wireshark evidence	Low	Provides monitoring baseline	Monitor for anomalies

No Critical or High finding is claimed because the evidence did not demonstrate a confirmed exploitable service or compromise. Medium ratings are used only where a missing host firewall or potential legacy plaintext service would materially increase exposure if present.

14. Security Hardening

Three defensive improvements were documented: enabling UFW, adding an explicit deny rule for TCP 23/Telnet, and adding an explicit deny rule for TCP 21/FTP. The final policy denies incoming traffic by default, allows outgoing traffic, and keeps logging enabled at low level.

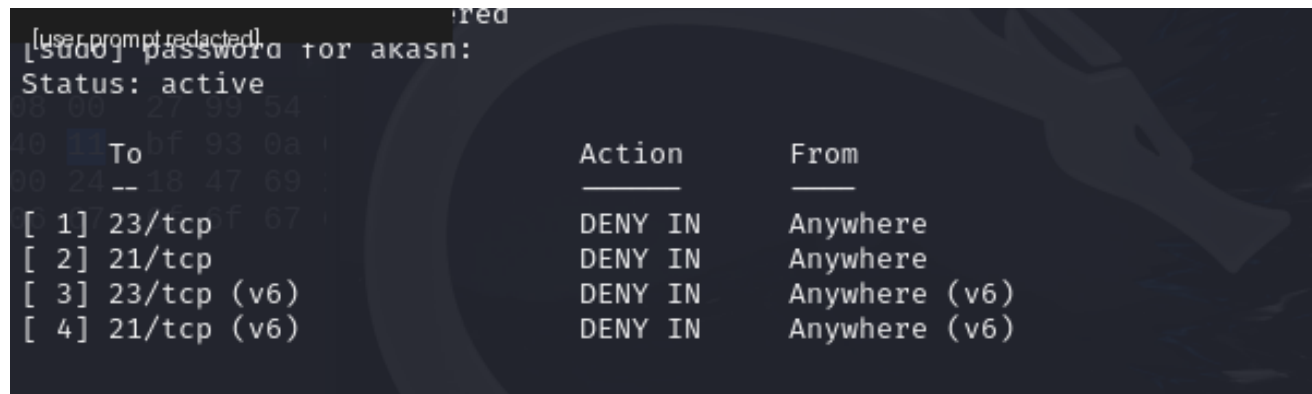


Figure 7. UFW numbered rules for TCP 21 and 23, including IPv6.

These explicit rules provide defense-in-depth if either legacy service is accidentally enabled later.

15. Before/After Comparison

Security Issue	Before	Action Taken	After	Improvement
Host firewall	UFW inactive	sudo ufw enable	UFW active; deny incoming	Host-based inbound filtering
Telnet control	No explicit 23/tcp deny	sudo ufw deny 23/tcp	IPv4/IPv6 DENY rules	Defense-in-depth
FTP control	No explicit 21/tcp deny	sudo ufw deny 21/tcp	IPv4/IPv6 DENY rules	Defense-in-depth

```
[user prompt redacted]
Nmap scan report for 10.0.2.15
Host is up (0.000088s latency).
PORT      STATE SERVICE
21/tcp    closed ftp
23/tcp    closed telnet
Nmap done: 1 IP address (1 host up) scanned in 0.15 seconds
```

Figure 8. Post-hardening targeted Nmap verification: TCP 21 and 23 closed.

```
UFW status verbose
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), disabled (routed)
New profiles: skip

To Action From
--  --  --
23/tcp DENY IN Anywhere
21/tcp DENY IN Anywhere
23/tcp (v6) DENY IN Anywhere (v6)
21/tcp (v6) DENY IN Anywhere (v6)
```

Figure 9. Final UFW verbose status.

The report does not claim that UFW changed ports 21/23 from open to closed, because earlier Nmap evidence did not show them open. The demonstrated improvement is the addition of explicit firewall controls and a deny-by-default inbound posture.

16. Security Recommendations

1. Keep UFW enabled and maintain a deny-by-default inbound policy. Review firewall rules periodically.
2. Avoid legacy plaintext protocols such as Telnet and traditional FTP. Use encrypted alternatives such as SSH/SFTP when required.
3. Perform periodic authorized Nmap scans and compare results with an approved baseline to detect unexpected services.
4. Monitor DNS, ICMP, and TCP patterns. Investigate anomalies using evidence rather than assuming unusual traffic is malicious.
5. Keep the operating system and installed packages updated and remove unnecessary services/software.

17. Challenges Faced

Traffic capture initially appeared blank when the wrong interface/filter combination was used. VirtualBox network-adapter configuration also required troubleshooting. Capturing on the Kali VM's active eth0 interface and generating controlled traffic made the analysis reliable.

Another challenge was interpreting results accurately: a closed port indicates no listening service and should not automatically be described as a firewall block.

18. Conclusion

The Week 3 assessment successfully combined network discovery, Nmap scanning, Wireshark analysis, security review, UFW hardening, and verification. Three active lab hosts were discovered, no exposed service was detected on the 1000 commonly scanned TCP ports of 10.0.2.15, normal ICMP and DNS traffic was analyzed, and defensive firewall controls were applied.

19. References

Reference categories specified by the assignment guideline: Nmap official documentation/reference guide and NSE documentation; Wireshark User's Guide and display-filter documentation; MITRE ATT&CK; CVE Program.

20. Screenshots / Evidence & Privacy Review

Evidence included in this report: network configuration, Nmap host discovery, Nmap service scan, ICMP traffic, DNS traffic, TCP SYN analysis, UFW numbered rules, targeted port verification, and final UFW status.

Privacy controls applied to this submission copy: personal names were not placed on the cover; document metadata uses the generic author 'Cybersecurity Intern'; screenshot title bars and terminal prompt identity were cropped/redacted where needed; no password, private key, API key, credential, email address, phone number, precise location, or public account identifier is included.

Private lab IP addresses such as 10.0.2.15 are retained because they are necessary technical evidence and are not publicly routable.

Before public GitHub submission, upload only reviewed project files. Do not upload raw terminal history, SSH keys, .env files, browser profiles, credential stores, access tokens, or unreviewed screenshots.

Final Assessment Summary

Network discovery: 3 active lab hosts. Service exposure: no listening service detected on the 1000 commonly scanned TCP ports of 10.0.2.15. Traffic observed: ICMP, DNS, TCP/HTTP-related test traffic. Hardening: UFW enabled with deny-incoming default plus explicit TCP 21/23 deny rules. Verification: TCP 21 and 23 closed; UFW active.

END OF REPORT